PURDUE UNIVERSITY | Elmore Family School of Electrical and Computer Engineering

ECE 49022 – Electrical Engineering Senior Design Projects
Fall 2026 – Subsystem-Level Engineering Design Requirements Report

MASHD: Multifunction Appliance for Smart Home Display
Written by: Javier Aldana

1.0 Team Member Assigned Subsystem(s)

Power System
User Display

2.0 List of your Product-Level Engineering Design Requirements

1) The product shall be powered over a cable coming from a wall outlet.
2) The product shall supply the necessary voltages at the necessary currents for every component.
3) The product shall be clearly visible from across the room to someone with 20-20 vision or corrective lenses.

3.0 List of your Subsystem/Component-Level Engineering Design Requirements

1.1) The power system will source all its power from a standard 120V, 60Hz power outlet.
2.1) **(TIDR)**The Power system shall be rated for 2A at 3.3V,13A at 5V, and 10mA at –3.3V. (This is above the necessary max current ratings)
2.2) (TIDR) The product shall have overcurrent protection for the 5V and 3.3V rails.
2.3) **(TIDR)** The product shall have current monitoring on the 5V and 3.3V rails.
3.1) The product shall have an electrical display that changes based on user's requests.

4.0 List of Stretch Requirement Statements for your Subsystem(s)

S1.1) The product should be powered off a USBC cable.
S2.1) (TIDR) The product should have all power converters above 90% efficiency.
S3.1) (TIDR) The product could set images beyond strings to the display.

All of the current and voltage specifications come from the datasheets on the components picked out. Another big requirements definer is useability by the consumer; this product is made to be as simple to hook up as possible and as convenient as can be. The power electronics should be completely behind the scenes, and should not be a point of failure under any conditions.

The User Display is for the convenience and use of the consumer, it should be visible under party conditions in a large room.

5.0 Test Plan for Each Engineering Design Requirement Statement for your Subsystem(s)

Number	Base-Product Design Requirement	Test/Verification Plan
1.1	The power system will source all its power from a standard 120V, 60Hz power outlet.	Observed, plug the module into the wall and observe if power is sourced properly.

2.1	**(TIDR)** The Power system shall be rated for 2A at 3.3V,13A at 5V, and 10mA at –3.3V.	Turn component on, measure current and voltage across shunts, confirm board functionality.
2.2	(**TIDR**) The product shall have overcurrent protection for the 5V and 3.3V rails.	Without having sensitive electronics hooked up downstream, set a small load from power to ground and observe system reaction on scope.
2.3	The product shall have current monitoring on the 5V and 3.3V rails.	Monitor current across shunt resistors with an Oscilloscope.
3.1	The product shall have an electrical display that changes based on user's requests.	Have a user interface where the user can write, draw, or set the display image, observe funcionality.
S1.1	The product should be powered off a USBC cable.	Observed, plug USBC into wall outlet and see if module turns on.
S2.1	**(TIDR)** The product should have all power converters above 90% efficiency.	Hook up Power supply to input, monitor input current and voltage. Measure output voltage and current with oscilloscope or multimeter, take the ratio.
S3.1	(TIDR) The product could set images beyond strings to the display.	Push images and video to user display from microcontroller, observe functionality.